BharatVoice Bench

A multilingual speech-recognition evaluation framework | Great Indian AI Internship Challenge

The Challenge

Speech recognition performance can vary across languages, speakers, and acoustic environments. BharatVoice Bench examines how Gnani Prisma v2.5 transcribes English, Hindi, and Hinglish speech when background noise is introduced under controlled conditions.

The Solution

BharatVoice Bench is an evaluation framework—not a new speech-recognition model. It combines recorded speech samples, controlled noise generation, Gnani Prisma v2.5 speech-to-text transcription, and Word Error Rate (WER) analysis in a reproducible workflow. A Flask dashboard presents aggregate metrics and individual recording results so users can inspect performance across test conditions.

How It Works

1. Record: Collect clean English, Hindi, and Hinglish speech from three speakers. **2. Add noise:** Generate crowd, fan, and traffic noise variants at 0, 10, and 20 dB SNR. **3. Transcribe:** Submit audio to Gnani Prisma v2.5 through the speech-to-text API. **4. Evaluate:** Compare recognized text with reference transcripts using Python and JiWER. **5. Explore:** Display results through a Flask-based web dashboard.

Evaluation Snapshot

81 noisy recordings generated	**78** successful noisy transcriptions	**5.45%** normalized corpus WER	**5.63%** mean per-record WER

Three additional Hindi crowd-noise requests returned HTTP 403 Cloudflare block responses; these were infrastructure-level failures, not ASR outputs, and were excluded from WER calculations. The normalized WER figures should be interpreted with the documented text-normalization rules and evaluation limitations.

Technology Used

Gnani: Prisma v2.5 speech-to-text API. **Implementation:** Python, Flask, FFmpeg, JiWER, HTML/CSS, and data visualizations.

Project Links

Source code: https://github.com/niniwritescode/BharatVoice-Bench
Live dashboard: https://bharatvoice-bench.onrender.com
Video demo: https://youtu.be/YbLFRq7ov_k

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